

* Features of Java + other languages

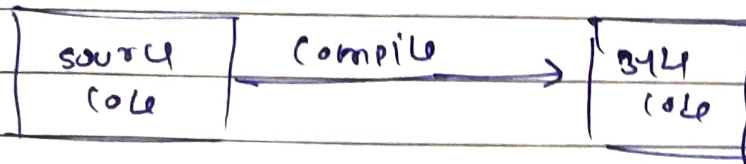
- ① simple : Java is Related to C to C++
 languages include : Basic syntax, variable, loops, variables, rule, if-else, for, while etc.
 C++ include object, class, polymorphism etc.

- ② pure oriented:
 - everything include in classes & objects.
 - without class we can not create java program.
 In java main() also included inside the class.

- ③ compile & interpreted (RUN)

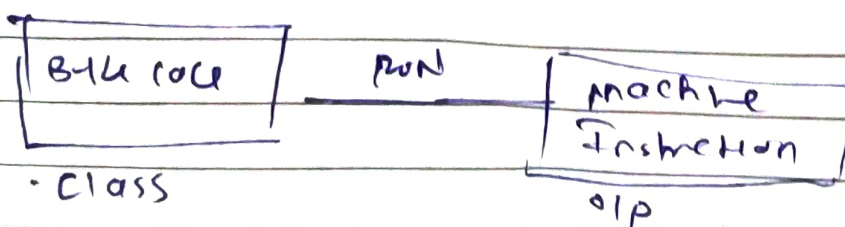
- ① compile : Java c tool is used to compile java program

syntax: javac filename.java



- a) interpreter (RUN)

syntax: Java classname
 e.g Java addition



- 4) portable:
 - 1 comp program easily run on another comp
 - comp. i.e. portable
 - In java once byte code file is generated then it can run on any comp
- 5) platform independent
 - java program run on any platform i.e. any OS
 - e.g. windows, Linux, DOS, Android
- 6) High performance
 - java support auto garbage collection
- 7) Dynamic :- (Native support)
- 8) Robust.
- 9) Multithreaded

A

```

class Add {
    public static void main (String arg[]) {
        System.out.println("welcome to Java");
    }
}

```

↑ ↑ ↓ ↑
 class object method arg.

- 1) public: Access specifier
- 2) static: keyword
static means globally without object.
- 3) void: return type of function
- 4) main(): Name of function
- 5) string array: when passing string array to main() then calling automatically

- To compile with package

```
import java.util
```

```
import java.util.*;
```

* check no is even or odd

```
class numbe
```

```
{
```

```
    public static void main (String args[])
```

```
    {
```

```
        int a=10;
```

```
        if (a%2 == 0)
```

```
            System.out.println("even");
```

```
        else
```

```
            System.out.println("odd");
```

```
    }
```

```
}
```

* Accept value from user

1) import util package

import java.util.*;

2) create object of scanner class

Scanner sc = new Scanner(System.in)

↓
Class Object memory
 allocate

connect to keyboard
cip device

3) Read values

int a = sc.nextInt();

float b = sc.nextFloat();

String s = sc.next();

* Java Editions

① J2SE (Java to Standard Edition)

- Fundamentals programming

- create desktop application, console application

② J2ME (Java to Micro Edition)

- Embedded system, mobile

- Battery, limited, small size, wireless device

③ J2EE (Java to Enterprise edition)

- Advanced java

- web, client-server, network application

* Object:

- object is the variable of class
- object is used to call member function of class
- when we create object of class then allocate memory to that class

Syntax:

class name objectname

Area Ob;

eg Area Ob = new Area();

* calling

Function

Syntax:

objectname . functionname (values);

eg ob: circle(2.345);